Jiyeol Oh

Daejeon, Republic of Korea | wlduf3187@kaist.ac.kr | acl.kaist.ac.kr | Google Scholar

Research Interests

Vehicle Dynamics and Control: Autonomous driving (risk assessment, path planning, path tracking); integrated chassis control (ABS, ESC, torque vectoring); parameter & state estimation; steady-state analysis (handling diagram, region of attraction estimation)

Control Theory: Nonlinear stability analysis (bifurcation theory); modal analysis (mode classification and tracking, modal controllability and observability); robust and adaptive control (LMIs, H-infinity control, disturbance observer)

Vehicle-Trailer: Axle load optimization; weight distribution hitch; trailer sway & swing detection and mitigation

Education

Ph.D. **KAIST**, Mechanical Engineering — Mar 2022 – Aug 2026
- Dissertation: Open-Loop System Optimization of Light Vehicle-Trailer Combinations for Improving Stability and Maneuverability
- Advisor: Prof. Seibum Choi

M.S. **KAIST**, Mechanical Engineering — Mar 2020 – Feb 2022
- Thesis: Torque Distribution Strategy of a In-Wheel-Motored Vehicle based on Tire-Vehicle Interactive Model
- Advisor: Prof. Seibum Choi

B.S. **KAIST**, Mechanical Engineering — Mar 2015 – Feb 2020

Publications

Optimal tongue weight selection criteria for light vehicle-trailer combinations — 2025
Jiyeol Oh, Seibum Choi
(Vehicle System Dynamics)

Operating principles of weight distribution hitches and their impact on handling characteristics of vehicle-trailer systems — 2026
Jiyeol Oh, Seibum Choi
(Vehicle System Dynamics)

Design of a Sensorless Trailer Anti-Lock Braking System Based on Trailer Swing Detection — 2026
Semin Jeong, Jiyeol Oh, Seibum Choi
(2026 American Control Conference (ACC))

Lateral Dynamics Modeling of General n-unit Combination Vehicles via Constraint Augmentation — 2026
Jiyeol Oh, Haewoo Lee, Seibum Choi
(2026 IEEE Conference on Decision and Control (CDC))

Efficient Collision Avoidance Trajectory Optimization for Autonomous Vehicles based on Multiple Time Constants Point Mass Model — 2026
Haewoo Lee, Jiyeol Oh, Seibum Choi
(Vehicle System Dynamics (under review))

Modal Family Classification for Vehicle-Trailer-Driver Lateral Dynamics 2026
Jiyeol Oh, Haewoo Lee, Seibum Choi
(2026 International Conference on Control, Automation and Systems (ICCAS) (under review))

Modal Analysis Framework for Vehicle-Trailer Lateral Dynamics: Mode Classification, Tracking, and Analysis 2026
Jiyeol Oh, Haewoo Lee, Jinhwan Jeon, Seibum Choi
(Vehicle System Dynamics (in progress))

Research Projects

Hyundai Motor Company, Project Lead — Mar 2024 – July 2026
- Project: Development of Control Algorithm for Enhanced Stability in Tow-Mode of xEV
- PI: Prof. Seibum Choi
- Contributions: Parameter & state estimation; vehicle-trailer stability index development; trailer sway and swing detection and mitigation

Ministry of Trade, Industry and Energy (MOTIE), Project Lead — Apr 2022 – present
- Project: Development of Lv.4+ Autonomous Driving Vehicle Platform for Point-to-Point Driving to Logistics Centers for Heavy Trucks
- PI: Prof. Seibum Choi
- Contributions: Parameter estimation; fault diagnosis and fault-tolerant control; Minimal Risk Maneuver (MRM) strategy development

Hyundai Motor Company, Research Assistant — Sept 2020 – Aug 2021
- Project: Driver-Customized Acceleration/Deceleration Pattern Derivation
- PI: Prof. Seibum Choi
- Contributions: Driver-specific feature extraction based on Discrete Wavelet Transform (DWT)

Skills

Software: MATLAB, Simulink, CarSim

Programming: Python

Languages: Korean (Native), English (Business), Japanese (Intermediate)